SLEEVEDRIFT · PRODUCT REQUIREMENTS AND SPECIFICATIONS

SleeveDrift product requirements and specifications

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-11-02	Amish Chadha	Kit 1.9.1 upgrade: renamed SVD-PRS-001 (was SVD-REQ-001); the twelve requirements R1 to R12 migrated to PR-001 to PR-012 with the old number in the Was column, each traced to a user requirement in SVD-URS-001; PR-013 (cost), PR-014 (workshop build) and PR-015 (signals) added from the user needs; no target changed; status of PR-013 recorded as over target
0.5	2026-10-04	Amish Chadha	R3 restated by Amish's round-3 decision 4A (SVD-DDR-004): moves loose spoil faster than the face produces it, at least 0.4 m3/h; met at 0.42 m3/h
0.4	2026-10-03	Amish Chadha	Status of R1, R3, R8 and R9 updated for Amish's decisions 24B, 25A and 26C of 2026-10-03 (SVD-DDR-003): portal drill drive, rope inspection and spare set, 600 mm stretch held for the co-design partner; targets unchanged

0.3	2026-10-03	Amish Chadha	TRL 3 status from SVD-CAL-001 on the constructable design (SVD-DDR-002); targets unchanged
0.2	2026-10-03	Amish Chadha	TRL 2 update; concept status for every requirement; R11 (overload limit) and R12 (gas monitoring) added under Amish's pre-approval (SVD-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold (as SVD-REQ-001)

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What the kit must do, each requirement traced to a user need in SVD-URS-001 (links in SVD-RTM-001). The requirements describe option A, the rope shuttle of sledges, which is the current design and is **proposed, awaiting Amish** as the choice among the three concept options (SVD-OPT-001). Of the fifteen, thirteen are met on paper or by design and one is managed by inspection and spares until the endurance trial (PR-009); **PR-013 (cost) is over the value-engineering target** at USD 10,804.00 of parts against USD 5,000. Targets are unchanged from the scaffold except PR-003, which Amish restated on 2026-10-04. The figures in the status column come from the calculation note SVD-CAL-001 and are estimates on paper, not test results; every requirement is verified by test at TRL 4 or later.

Amish, 2026-10-03: "i agree with all the 46 recommendations you provided. please proceed." For SleeveDrift that decided three items as recommended (SVD-DDR-003): a portal drill drive on each winch for the haul rate (24B), rope inspection every shift with a spare set for shift durability (25A), and the 600 mm stretch of PR-001 held until the co-design partner answers (26C). On 2026-10-04 Amish decided option 4A: "For round 3, I agree with all your proposed recommendations". PR-003 is restated as "moves loose spoil faster than the face produces it; at least 0.4 m3/h" (hand-dug faces make about 0.27 m3/h) and is met at 0.42 m3/h (SVD-DDR-004).

> **Safety:** These requirements describe equipment used by people inside a rescue pipe under collapse debris. Meeting them on paper does not make the kit safe to use; it is published as an open engineering reference, not certified rescue, mining or confined-space equipment.

Table 1. Product requirements and specifications. Status at TRL 3 from SVD-CAL-001 (python docs/04-calcs/sizing.py).

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-007	Physical architecture	Fits the pipe	Every part passes through and assembles inside an 800 mm nominal pipe; hood and ring variants for 600 mm are a stretch	776 circle in a 780 bore	max	mm	Analysis, SVD-CAL-001 A (TRL 3); fit trial in pipe sections (TRL 4)	Met on paper for 800 mm: every piece passes a 776 mm circle in a 780 mm bore. 600 mm stretch open: held until the co-design partner says whether 600 mm pipes are used for rescue (decision 26C)	R1
PR-002	UR-001	Safety	Hood protection	The hood carries a 5 kN (1,124 lbf) distributed load with no contact with a test head form	5	min	kN	Analysis, SVD-CAL-001 B (TRL 3); load test on CalRig (TRL 4)	Met on paper: 5.5 MPa in the crown, tip deflection under 0.1 mm	R2
PR-003	UR-002	Performance	Spoil haulage	Moves loose spoil faster than the face produces it; at least 0.4 m3/h (14 ft3/h), moved 60 m (197 ft) by a two-person portal crew (restated 2026-10-04, decision 4A; hand-dug faces make	0.4	min	m3/h	Analysis, SVD-CAL-001 D (TRL 3); timed trial in a 60 m pipe run (TRL 4)	Met as restated: about 0.42 m3/h with a right-angle drill driving each portal winch (decision 24B), a margin of 5 %; 0.22 m3/h by hand as the fallback, which is	R3

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				about 0.27 m3/h)					below the face rate	
PR-004	UR-003	Usability	Haul force	Portal crank force under 200 N (45 lbf) with loaded sledges over 60 m	200	max	N	Analysis, SVD-CAL-001 C (TRL 3); force gauge during trial (TRL 4)	Met on paper: 93 N at the start in high gear, 30 N in low gear	R4
PR-005	UR-004	Safety	Fresh air at the face	At least 1 m3/min (35 ft3/min) of fresh air delivered at the face over 60 m of duct (target, to be checked against confined-space guidance)	1	min	m3/min	Analysis, SVD-CAL-001 F (TRL 3); anemometer at duct outlet (TRL 4)	Met on paper: about 15 m3/min at the face	R5
PR-006	UR-006	Performance	Casualty evacuation	A 100 kg (220 lb) manikin moved 60 m through the pipe in under 5 minutes	5	max	min	Analysis, SVD-CAL-001 E (TRL 3); timed drill (TRL 4)	Met on paper: about 3.5 min with three people hauling hand over hand	R6
PR-007	UR-008	Deployment and assembly	Rapid set-up	Kit installed and hauling within 2 hours of arriving at the portal	2	max	h	Analysis, SVD-CAL-001 G (TRL 3); timed deployment exercise (TRL 4)	Met on paper: about 1.6 h estimated	R7
PR-008	UR-009	Electrical	No power in the pipe	Zero electrically powered items inside the pipe except battery lamps and	0	max	powered items	Design review (TRL 3)	Met by design: battery lamps, gas monitors (alarm devices) and sound-	R8

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				signal devices					powered telephones only; the drill drive stays at the portal	
PR-009	UR-010	Materials	Shift durability	72 hours of continuous operation without failure of rope, pulleys or sledges	72	min	h	Analysis, SVD-CAL-001 H (TRL 3); endurance trial (TRL 4)	Managed: rope abrasion over about 250 round trips at the drill rate cannot be shown on paper; both ropes are inspected against a wear gauge every shift and a spare set is carried (decision 25A). The endurance trial decides whether a tougher rope is needed	R9
PR-010	UR-011	Packaging and transport	Transportable	Whole kit in cases no heavier than 40 kg (88 lb) each, moveable by road and by hand to the portal	40	max	kg	Analysis, SVD-CAL-001 I (TRL 3); weighing and handling trial (TRL 4)	Met, thin: heaviest package 37.7 kg (the sledge bundle)	R10
PR-011	UR-012	Safety	Overload limit	Rope tension limited to 2.5 kN (562 lbf) by a release at each end of the train	2.5	max	kN	Analysis, SVD-CAL-001 C and D6 (TRL 3); release test of the	Met by design: rope factor 6.5 on the release load; drill	R11

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				or stretcher; every part sized on that limit				swivel pins and proof load on CalRig (TRL 4)	clutch slips at 1.55 kN in high gear	
PR-012	UR-005	Safety	Air monitoring	Oxygen, carbon monoxide, hydrogen sulphide and flammable gas monitored continuously at the face and at the portal, with alarms	4 gases, 2 places	min	gases, places	Design review (TRL 3); bump test before each shift (TRL 4)	Met by design: two four-gas monitors in the kit	R12
PR-013	UR-013	Cost	Affordable to hold	Parts cost of the priced bill of materials within the USD 5,000 value-engineering target (a control target, not a limit; Amish kept it on 2026-10-03, SVD-DDR-001 item 13)	5,000	max	USD	Priced BOM, bom/bom.csv; SVD-CAL-001 J	Over target: USD 10,804.00 in parts, USD 5,804.00 over. The two winches (USD 2,300), four-gas monitors (USD 1,100), signal line (USD 860), duct (USD 765) and blower (USD 750) are the largest lines. Proposed, awaiting Amish: hold the target, or restate it	none

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
									for a rescue kit	
PR-014	UR-014	Materials	Made by a general workshop	Every made part shall be cut, rolled, drilled, bent or welded from stock steel section, bar and sheet with workshop tools; bought parts limited to catalogue items (winches, sheave, ropes, drills, blower, duct, monitors, fasteners); no casting and no machining beyond drilling and facing	Pass or fail	Pass or fail	n/a	Inspection of the BOM, the model and the build plan SVD-BLD-001 (TRL 3)	Met by design: the fidelity check lists every made part (cad/fidelity.json)	none
PR-015	UR-015	Electrical	Face and portal signals	A sound-powered telephone pair with a pull-cord bell as a mechanical backup runs between the face and the portal, with no battery in the pipe	1 pair, 1 backup	min	sets	Design review (TRL 3), SVD-DDR-001 item 8; call test (TRL 4)	Met by design: signal line, BOM line 24	none

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- The rescue pipe is steel of 800 mm outside diameter with a 10 mm wall (780 mm bore), as at Silkyara; the fit trial confirms the bore and the weld beads.
- Experienced hand miners or trained rescue crews will be available to use the kit.
- A blower and some power are available at the portal; nothing inside the pipe uses mains power.
- The pipe is pushed forward by other means while crews dig; SleeveDrift does not push pipe.
- The pipe is horizontal or on a slight grade, so the train does not run away when a winch is eased.
- Targets are proposals for review with a co-design partner (SVD-PRB-001). PR-013 to PR-015 are new at the kit 1.9.1 upgrade and are proposed, awaiting Amish, with the rest.